INCH POUND

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DETAIL SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, M27 REMANUFACTURED METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M27 Cartridge Actuated Initiator Remanufactured Metal Parts Assembly (see 6.1)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-2550	Ammunition, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-413	Visual Inspection Guide for Elastomeric O-rings

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/

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AMSC N/A FSC 1377

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

NAVAL ORDNANCE STATION, INDIAN HEAD (GAGE CODE 19200 and 19203)

10520551	Initiator, Cartridge Actuated, M27, Metal Parts Assembly
8830984	Box, Fiberboard (inner) packing, Ammunition for Initiator, Cartridge Actuated M27, M86, M87, M99, or XM107
8830985	Box, Packing, Ammunition, for Initiator, Cartridge Actuated M27, M86, M87, M98, M99, or XM017
9241155	Box, Fiberboard (Outer) Alternative Packing, Ammunition, for Initiator, Cartridge Actuated M27, M86, M87, M98, M99, or XM107

(Application for copies should be addressed to Commanding Officer, Naval Surface Warfare Center, Indian Head, Technical Information Branch (Code 4230), Indian Head, MD 20640.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY

ASQ A8402	Quality Management and Quality Assurance – Vocabulary
ASQ Z1.4	Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the American Society for Quality Control, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the initiators shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure there from has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal. The remanufactured initiator metal parts assembly shall contain a new seal, initiator spring, O-ring, and safety pin.

3.2 Hydrostatic pressure. The initiator chamber and cap shall withstand a hydrostatic pressure of 14,00 to 15,00 pounds per square inch (psi) for 15 seconds, minimum, without deformation or mechanical failure (see 6.3.1).

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3.3 Firing pin release. The firing pin of the metal parts assembly, less safety pin, shall release from the cocked position when the application of an axial load of 15 to 30 pounds has withdrawn the initiator pin 0.370 inch, minimum, and 0.425 inch, maximum.

3.4 Initiator pin security. Application of an axial load of 200 to 220 pounds to the initiator pin of the metal parts assembly shall not withdraw the initiator pin past the safety pin. Load application shall cause no permanent deformation of the safety pin.

3.5 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component, regardless of configuration. Remanufactured exterior components shall have the existing protective finish stripped and the specified protective finish reapplied (see 6.6).

3.6 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using procedures in 4.6.5 (see 6.5).

3.7 Metal parts assembly. Parts shall be combined to produce an assembly, which conforms to the requirements of the metals parts assembly drawing. All threaded joints shall be assembled hand tight (see 6.5).

3.8 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following (see 6.6):

3.8.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.8.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Conformance inspection.

4.1.1 Quality assurance terms and definitions. Reference shall be made to ASQ A8402 to define the quality assurance terms used.

4.1.2 Inspection. Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection conditioning. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 20 to 25 °C (68 to 77 °F)
b. Altitude: Normal ground
c. Humidity: 50 percent relative, maximum

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d. Atmospheric pressure: 14.7 psia.

4.3 Quality conformance inspection.

4.3.1 Lot.

4.3.1.1 Submission of product. The product shall be submitted in accordance with ASQ Z1.4.

4.3.1.2 Size of lot. For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 assemblies, maximum, provided the homogeneity requirements of inspection lots have been retained.

4.3.2 Examination. Of reprocessed parts 100 percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.3.2.1 Examination of reclaimed parts. Examination of reuseable parts, which require no reprocessing shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be reprocessed or rejected.

4.3.2.2 Chamber, Initiator, Drawing 10520407, covering a detail of Drawing 10520551.

Characteristic	Defect	Method of Inspection
Critical		
None Defined		
Major		
101	Pitch diameter of large external thread, min	Gage
102	Major diameter of large external thread, min	Gage
103	Pitch diameter of small external thread, min	Gage
104	Major diameter of small external thread, min	Gage
105	Pitch diameter of internal thread, max	Gage
106	Minor diameter of internal thread, max	Gage
107	Thread damage	Visual
108	Metal defective	Visual
109	Protective finish, inadequate or defective	Visual
110	Surface finish of cavity and angle of gas port, improper	Visual
111	Burr	Visual
112	Foreign matter present	Visual
Minor		
201	Surface finish, improper, except as otherwise classified	Visual

4.3.2.3 Cap, Initiator, Drawing 10520552, covering a detail of Drawing 10520551.

Critical		
1	Metal defective in mounting surface of flange	Visual

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Major		
101	Pitch diameter of small internal thread, max	Gage
102	Minor diameter of small internal thread, max	Gage
103	Pitch diameter of large external thread, max	Gage
104	Minor diameter of large external thread, max	Gage
105	Metal defective except as otherwise classified	Visual
106	Protective finish, inadequate or defective	Visual
107	Burr	Visual
108	Foreign matter present	Visual
Minor		
201	Surface finish, improper	Visual

4.3.2.4 Pin, Initiator, Drawing 10520553, covering a detail of Drawing 10520551.

Critical
None Defined

Major		
101	Metal defective	Visual
102	Protective finish, inadequate or defective	Visual
103	Surface finish of groove in head and face of shoulder, improper	Visual
104	Burr	Visual
105	Foreign matter present	Visual
Minor		
201	Surface finish, improper, except as otherwise classified	Visual

4.3.2.5 Housing, Firing Pin, Drawing 10520410, covering a detail of Drawing 10520551.

Critical
None defined

Major		
101	Thread damaged	Visual
102	Metal defective	Visual
103	Protective finish, inadequate or defective	Visual
104	Surface finish of cavity, improper	Visual
105	Burr	Visual
106	Foreign matter present	Visual
Minor		
201	Pitch diameter for thread, min	Visual
202	Major diameter for thread, min	Visual

4.3.2.6 Pin, Firing, Drawing 10520411, covering a detail of Drawing 10520551.

Critical
None Defined

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Major

101	Metal defective	Visual
102	Protective finish, inadequate or defective	Visual
103	Surface finish of body, improper	Visual
104	Burr	Visual
105	Foreign matter present	Visual

Minor

None defined

4.3.2.7 Filter, Drawing 10520414, covering a detail of Drawing 10520551.

Critical

None Defined

Major

101	Pitch diameter of thread, min	Gage
102	Major diameter of thread, min	Gage
103	Thread damaged	Visual
104	Nylon patch not properly located or fused	Manual/Visual
105	Metal defective	Visual
106	Protective finish, inadequate or defective	Visual
107	Burr	Visual
108	Foreign matter present	Visual

Minor

None defined

4.3.2.8 Assembly, Drawing 10520551, less all drawings except 10520410.

Critical

1	Firing pin protrusion	Gage
2	Steel ball, missing	Gage
3	O-ring missing	Visual
4	Seal missing	Visual

Major

101	Burr	Visual
102	Foreign matter present	Visual

Minor

None defined

4.3.5.9 Assembly, Drawing 10520551.

Critical

None defined

Major

101	Firing pin cocked	Manual
102	Initiator pin not free to rotate	Manual
103	Any part missing, except as otherwise classified	Visual
104	Improper assembly	Visual
105	O-ring not lubricated	Visual
107	Marking missing, incorrect or illegible	Visual

Minor

201	Protective cap missing	Visual

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202	Protective finish inadequate or defective	Visual

4.3.2.10 Box, Fiberboard (inner Ammunition Packing, Drawing 8830984.

Critical
None defined
Major
None defined
Minor

201	Contents loose	Manual
202	Box damaged	Visual
203	Sealing strip missing, incorrectly applied	Visual
204	Marking missing, incorrect or illegible	Visual
205	Cut damaged or improperly sealed bag	Visual

4.3.2.11 Box, Ammunition, Packing, Drawing 8830985.

Critical
None defined
Major
None defined
Minor

201	Contents loose	Manual
202	Strapping missing, loose, broken, or improperly assembled	Manual/Visual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect or illegible	Visual
206	Number of units incorrect	Visual

4.5.2.12 Box, Fiberboard (Outer) Alternative, Ammunition Packing, Drawing 9241155.

Critical
None defined
Major
None defined
Minor

201	Contents loose	Manual
202	Box broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect or illegible	Visual
205	Tape incomplete	Visual

4.3.2.13 Spring, Initiator, Drawing 10520412, covering a detail of Drawing 10520551.

Critical
None defined
Major

101	Load at specified height	Gage
102	Closed length, max	Gage
103	Ends incorrect	Visual

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104	Protective finish, inadequate or defective	Visual
105	Burr	Visual
106	Foreign matter present	Visual
Minor		
201	Diameter outside	Gage
202	Free length	Gage

4.3.2.14 Pin, Safety, Drawing 10522817, covering a detail of Drawing 10520551.

Critical		
None defined		
Major		
101	Diameter of wire	Gage
102	Profile, incorrect	Visual
103	Metal defective	Visual
104	Protective finish, inadequate or defective	Visual
105	Burr	Visual
106	Foreign matter present	Visual
Minor		
201	Diameter of head, outside	Measure
202	Length between high and low ends	Measure
203	Total length	Measure
204	Location of center loop	Measure
205	Radius of center loop, incorrect	Measure

4.3.2.15 Seal Drawing 10520415, covering a detail of Drawing 10520551.

Critical		
None defined		
Major		
101	Diameter, outside, min	Gage
102	Width, min	Gage
103	Thickness, min	Gage
104	Cut, cracked or split	Visual
105	Flash	Visual
106	Foreign matter present	Visual
Minor		
201	Diameter, outside, max	Gage
202	Width, max	Gage
203	Thickness, max	Gage

4.4 Inspection testing. The following tests shall be performed in accordance with the provision of 4.5.

4.4.1 Hydrostatic pressure. Each chamber and cap shall be subjected to the hydrostatic pressure test. Failure of any cap or chamber subassembly to comply with the requirements of 3.2 shall be cause for rejection of the subassembly.

4.4.2 Firing pin release. The initiator pin of each partial assembly consisting of all parts except the safety pin and chamber, shall be subjected to the firing pin release test. Failure of any partial assembly to comply with the requirements of 3.3 shall be cause for rejection of the subassembly.

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4.4.3 Initiator pin security. The initiator pin of each subassembly consisting of all parts except the chamber shall be subjected to the initiator pin security test. Failure of any subassembly to comply with the requirements of 3.4 shall be cause for rejection of the subassembly.

4.4.4 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.4.5 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any unit in the inspection sample to comply with the requirement of 3.6 shall be cause for rejection of the assembly.

4.5 Test methods and procedures.

4.5.1 Hydrostatic pressure. Prior to the application of protective finish, the cap and chamber shall be mounted in a device, using a fixture for properly sealing all openings and subjected to the internal hydrostatic pressure to determine compliance with the requirements of 3.2.

4.5.2 Firing pin release. The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture, which registers the amount of pull.

4.5.3 Initiator pin security. The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with a fixture, which registers the amount of pull. The test shall be performed with the safety pin in both positions.

4.5.4 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the component drawing.

4.5.5 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater the 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90-degree rotation of the assembly for 4 full passes.

4.6 Inspection of packaging. The sampling and inspection of the packing and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contractor order (see 6.2). When actual packaging of material needs to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense

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Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may helpful but is not mandatory.)

6.1 Intended use. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems. The assembly was designed for use in military aircraft and has no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. The responsible for inspection activity (see 4.1)
d. Any special inspection conditions (see 4.3)
e. Level of marking and packing (see 5.1 and 6.3)

6.3 Contract packaging, packing and marking. The following requirements should be specified in the contract.

6.3.1 Packing and marking level. Packing and marking should be Level A or B, as specified by the contracting activity.

6.3 .1.1 Level A. Level A should be in accordance with Drawings 8830984 and 8830985 or SPI F00-138-3770.

6.3.1.2 Level B. Level B should be in accordance with Drawings 8830984 and 9241155 or SPI F00-138-3770.

6.3.1.3 Lot numbers. Lot numbers should be in accordance with MIL-STD-1168.

6.4 Definitions.

6.4.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5 Metal parts assembly. When remanufacturing and loading of assembly is performed in an integrated plant the metal parts assembly requirement may be waived and the residual magnetism requirement performed on the loaded assembly only.

6. 6 Exterior components. Remanufactured exterior components are as follows: Cap, Initiator Pin, and Chamber.

6.7 Subject term (key word) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew

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Escape systems, aircrew
Life-saving systems, aircrew

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes. These changes were authorized by Indian Head ECP 045220I018.

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Preparing activity
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